Span Categories & their Uses IV: Universal Biadjointable Functors

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2nd of February of 2026

A **3-functor formalism** is a lax symmetric monoidal $(\infty,2)$-functor

$$D : \mathrm{Span}(C,E) \longrightarrow \mathrm{Cat}_\infty$$

For any $f : X \to Y$ in C, define $f^* := D(Y \xleftarrow{f} X = X) : D(Y) \longrightarrow D(X)$

For any $f : X \to Y$ in E, define $f_! := D(X = X \xrightarrow{f} Y) : D(X) \longrightarrow D(Y)$

A **6-functor formalism** is a 3-functor formalism for which the functors $- \otimes A$, f^* and $f_!$ admit right adjoints

Difficult to construct! Solution: Construct the pullback functor

$$(-)^* : C^{op} \longrightarrow \mathrm{Cat}_\infty \quad \text{and extend it to } \mathrm{Span}(C,E)$$

One extension: Assume E decomposes in two classes I and P

$$\left.\begin{array}{l} \text{For } i \in I,\ i^* \text{ has left adjoint } i_\# \\ \text{For } p \in P,\ p^* \text{ has right adjoint } p_* \end{array}\right\} \Longrightarrow (-)_! := p_* i_\# : C \longrightarrow \mathrm{Cat}_\infty$$

Generalization: $\mathbb{D}$ $(\infty,2)$-category instead of Cat_∞

$\mathrm{SPAN}_2(C,E)_{P,I} :=$ $(\infty,2)$-category of spans, with spans of spans as 2-morphisms

Theorem A. C ∞-cat, $I, P \subset E \subset C$ wide subcats s.t.

(i) I and P are closed under base change and left cancellable

(ii) Every morphism in E factors as $p \circ i$ with $i \in I$ and $p \in P$

(iii) Every map in $I \cap P$ is truncated

Then the inclusion $h : C^{op} \hookrightarrow \mathrm{SPAN}_2(C,E)_{P,I}$ induces an equivalence

$$h^* : \mathrm{FUN}(\mathrm{SPAN}_2(C,E)_{P,I}, \mathbb{D}) \xrightarrow{\sim} \mathrm{FUN}_{(I,P)-\mathrm{biadj}}(C^{op}, \mathbb{D})$$

$I = P = E$ and $(2,2)$-categories
Balmer and Dell'Ambrogio

(I,P)-biadjointable functors and nat. trans.

Intuition. Let $m \leq n \leq \infty$, an (n,m)-category is a higher category having i-morphisms for $i \leq n$ which are invertible $i > m$.

Notation. categories $:=$ $(\infty,1)$-categories
n-categories $:= (\infty, n)$-categories

FUN, HOM categories in a 2-categorical setting

CAT 2-category of categories, CAT_2 3-category of 2-categories

Goal: Given a category C and suitable subcategories I and P:

- Introduce (I,P)-biadjointable functors
- The universal biadjointable functor
- Recognition principle $\rightsquigarrow$ Used to prove Theorem A in future talks

$\mathbb{D}$ 2-category, a commutative square $\begin{tikzcd} X' \arrow[r, "g^*"] \arrow[d, "f^*"'] & X \arrow[d, "k^*"] \\ Y' \arrow[r, "h^*"'] & Y \end{tikzcd}$ in $\mathbb{D}$ is:

(1) **Vertically left adjointable** if f^* and k^* admit left adjoints $f_\#$ and $k_\#$ and the Beck-Chevalley map

$$BC_\# : k_\# h^* \longrightarrow k_\# h^* f^* f_\# \simeq k_\# k^* g^* f_\# \longrightarrow g^* f_\#$$ is an equivalence

(2) **Vertically right adjointable** " admit right adjoints f_* and h_*

" $BC_* : g^* f_* \longrightarrow k_* h^*$ "

(3) **Horizontally left/right adjointable** same but for g^* and h^*

(4) **Biadjointable** if it is horizontal left adjointable and $\begin{tikzcd} X' \arrow[d, "f^*"] & X \arrow[l, "g_\#"'] \arrow[d, "k^*"] \\ Y' & Y \arrow[l, "h_\#"] \end{tikzcd}$ is vertically right adjointable

$\mathcal{C}$ category, $I, P \subseteq \mathcal{C}$ wide subcats closed under base change, $\mathbb{D}$ 2-category and $F : \mathcal{C}^{op} \to \mathbb{D}$ denote $f^* := F(f) : F(Y) \to F(X)$ $\forall f : X \to Y$ in $\mathcal{C}$

① F is **left I-adjointable** if

$$\forall \text{ pullback } \begin{tikzcd} X' \arrow[r, "j"] \arrow[d, "g"'] \arrow[dr, phantom, "\lrcorner", very near start] & X \arrow[d, "f"] \\ Y' \arrow[r, "i"'] & Y \end{tikzcd} \quad i \in I \rightsquigarrow \begin{tikzcd} F(Y) \arrow[r, "i^*"'] \arrow[d, "f^*"] & F(Y') \arrow[d, "g^*"] \\ F(X) \arrow[r, "j^*"'] & F(X') \end{tikzcd} \text{ is horizontally left adjointable}$$

② Dually, F is **right P-adjointable** if

$$\forall \text{ pullback } \begin{tikzcd} X' \arrow[r, "g"] \arrow[d, "q"'] \arrow[dr, phantom, "\lrcorner", very near start] & X \arrow[d, "p"] \\ Y' \arrow[r, "f"'] & Y \end{tikzcd} \quad p \in P \rightsquigarrow \begin{tikzcd} F(Y) \arrow[r, "f^*"] \arrow[d, "p^*"] & F(Y') \arrow[d, "q^*"] \\ F(X) \arrow[r, "g^*"'] & F(X') \end{tikzcd} \text{ is vertically right adjointable}$$

③ F is **(I,P)-biadjointable** if it is left I-adjointable, right P-adjointable and

$$\forall \text{ pullback } \begin{tikzcd} X' \arrow[r, "j"] \arrow[d, "q"'] \arrow[dr, phantom, "\lrcorner", very near start] & X \arrow[d, "p"] \\ Y' \arrow[r, "i"'] & Y \end{tikzcd} \quad \begin{matrix} p \in P \\ i \in I \end{matrix} \rightsquigarrow \begin{tikzcd} F(Y) \arrow[r, "i^*"'] \arrow[d, "p^*"] & F(Y') \arrow[d, "q^*"] \\ F(X) \arrow[r, "j^*"'] & F(X') \end{tikzcd} \text{ is biadjointable}$$

$\mathcal{C}$ category, $\mathcal{I}, \mathcal{P} \subseteq \mathcal{C}$ wide subcats closed under base change, $\mathbb{D}$ 2-category

$F, G : \mathcal{C}^{op} \longrightarrow \mathbb{D}$ and $\alpha : F \Rightarrow G$ natural transformation

$\alpha : F \Rightarrow G$ is **left $\mathcal{I}$-adjointable** if

$$\begin{tikzcd} F(Y) \arrow[r, "j^*"] \arrow[d, "\alpha_Y"] & F(X) \arrow[d, "\alpha_X"] \\ G(Y) \arrow[r, "j^*"] & G(X) \end{tikzcd}$$

is horizontally left adjointable

Same for **right $\mathcal{P}$-adjointable**

and **$(\mathcal{I}, \mathcal{P})$-biadjointable** = left $\mathcal{I}$-adjointable + right $\mathcal{P}$-adjointable

$\mathrm{FUN}_{(\mathcal{I},\mathcal{P})\text{-}\mathrm{badj}}(\mathcal{C}^{op}, \mathbb{D})$:= locally full sub-2-category of $\mathrm{FUN}(\mathcal{C}^{op}, \mathbb{D})$ spanned by $(\mathcal{I}, \mathcal{P})$-biadjointable functors and nat. transformations

Remarks.
- If $\mathcal{P}$ (resp. $\mathcal{I}$) consists only of the equivalences in $\mathcal{C}$ $(\mathcal{I}, \mathcal{P})$-biadjoint. = left $\mathcal{I}$-adjoint. (resp. right $\mathcal{P}$-adjoint)
- Any 2-functor preserves (bi)adjointable squares

C category, $I, P \subseteq C$ wide subcats closed under base change

An (I,P)-biadjointable functor $F : C^{op} \to \mathbb{B}$ is universal if the restriction

$$F^* : \mathrm{FUN}(\mathbb{B}, \mathbb{D}) \longrightarrow \mathrm{FUN}_{(I,P)\text{-biadj}}(C^{op}, \mathbb{D})$$

is an equivalence of 2-cats for every 2-cat $\mathbb{D}$.

$$\begin{tikzcd} C^{op} \arrow[r, "G"] \arrow[d, "F"'] & \mathbb{D} \\ \mathbb{B} \arrow[ur, dashed, "\exists \bar{G}"'] & \end{tikzcd}$$

Theorem (Existence). $\exists\, h : C^{op} \longrightarrow \mathbb{B}_{I,P}(C)$ universal

An (I,P)-biadjointable functor $F : C^{op} \to \mathrm{CAT}$ is free on a class $1_a \in F(a)$ with $a \in C$ if $\forall$ (I,P)-biadjointable functor $G : C^{op} \to \mathrm{CAT}$ the evaluation at 1_a $\mathrm{HOM}(F, G) \xrightarrow{\sim} G(a)$ is an equivalence

Theorem (Recognition principle). An (I,P)-biadjointable functor $F : C^{op} \to \mathbb{B}$ is universal $\iff$ it is essentially surjective and for all $a \in C$ $\mathrm{HOM}(F(a), F(-)) : C^{op} \longrightarrow \mathrm{CAT}$ is a free (I,P)-biadjointable functor on $\mathrm{id}_{F(a)}$

PROOF OF THE EXISTENCE OF THE UNIVERSAL BADJ FUNCTOR

Define a 3-cat BADJDATA$_2$ with the following data:

(1) Objects $(\mathbb{C}, \mathbb{Q}_L, \mathbb{Q}_R, \mathbb{Q}_B)$ where $\mathbb{C}$ is a 2-cat, $\mathbb{Q}_\bullet \subset \mathrm{FUN}([1]\times[1], \mathbb{C})$ full subcategories s.t. $\{\text{equivalences}\} \subset \mathbb{Q}_\bullet$ and $\mathbb{Q}_B \subset \mathbb{Q}_L \cap \mathbb{Q}_R$

Denote $\mathbb{L}$ the horizontal maps in squares of $\mathbb{Q}_L$

and $\mathbb{R}$ " vertical " " " " $\mathbb{Q}_R$

(2) Morphisms are 2-functors $F : \mathbb{C} \to \mathbb{C}'$ satisfying $F(\mathbb{Q}_\bullet) \subset \mathbb{Q}'_\bullet$

(3) 2-cells are nat. transf. $\alpha : F \Rightarrow G$ s.t. $\forall l : X \to Y$ in $\mathbb{L}$

$$\begin{tikzcd} F(X) \arrow[r, "F(l)"] \arrow[d, "\alpha_X"'] & F(Y) \arrow[d, "\alpha_Y"] \\ G(X) \arrow[r, "G(l)"] & G(Y) \end{tikzcd}$$

belongs to $\mathbb{Q}_L$, dually for $\mathbb{R}$ and $\mathbb{Q}_R$

(4) No condition on higher cells

PROOF OF THE EXISTENCE OF THE UNIVERSAL BADJ FUNCTOR

Ex. C category with $I, P \subseteq C$ wide subcats closed under base change

$j(C, I, P) := (C^{op}, \mathbb{Q}_L, \mathbb{Q}_R, \mathbb{Q}_B) \in \mathrm{BADJDATA}_2$ where

$\mathbb{Q}_L$ = pullbacks with horizontal maps in I

$\mathbb{Q}_R$ = pullbacks with vertical maps in P

$\mathbb{Q}_B = \mathbb{Q}_L \cap \mathbb{Q}_R$

Ex. Any 2-cat $\mathbb{D}$ determines $(\mathbb{D}, \mathbb{Q}_L, \mathbb{Q}_R, \mathbb{Q}_B) \in \mathrm{BADJDATA}_2$ where

$\mathbb{Q}_L$ = horizontally left adjointable squares

$\mathbb{Q}_R$ = vertically right adjointable squares

$\mathbb{Q}_B$ = biadjointable squares

Because 2-functors and 2-natural transformations preserve adjointability, this assignment defines a 3-functor

$$(-)_{badj} : \mathrm{CAT}_2 \longrightarrow \mathrm{BADJDATA}_2$$

Proof of the Existence of the Universal BAdj Functor

For every (C, I, P) as before and 2-cat $\mathbb{D}$, the forgetful map induces an equivalence

$$\mathrm{HOM}_{\mathrm{BADJDATA}_2}(j(C,I,P), \mathbb{D}_{badj}) \overset{\gamma}{\xrightarrow{\sim}} \mathrm{FUN}_{(I,P)\text{-}badj}(C^{op}, \mathbb{D}) \subseteq \mathrm{FUN}(C^{op}, \mathbb{D})$$

Lemma 1. $(-)_{badj}$ admits a left adjoint 3-functor $\hat{\mathbb{B}}$

Assuming Lemma 1, then

$$\mathrm{HOM}_{\mathrm{CAT}_2}(\hat{\mathbb{B}} j(C,I,P), \mathbb{D}) \overset{\varphi}{\xrightarrow{\sim}} \mathrm{HOM}_{\mathrm{BADJDATA}_2}(j(C,I,P), \mathbb{D}_{badj})$$

Then, there exists $\mathbb{B}_{I,P}(C) := \hat{\mathbb{B}} j(C, I, P)$ and $h := \gamma \circ \varphi(\mathrm{id})$ where h is universal and (I,P)-beadjointable by construction.

$\square$

Idea of the Proof of the Lemma 1

Define the **walking adjunction** ADJ is the 2-category with two morphism $l : \ominus \to \oplus$ and $r : \oplus \to \ominus$ s.t. $l \dashv r$.

Squares in $\mathbb{D}$

Lemma. (1) $(\mathrm{id} \times r)^* : \mathrm{FUN}([1] \times \mathrm{ADJ}, \mathbb{D}) \to \mathrm{FUN}([1] \times [1], \mathbb{D})$ is a locally full inclusion with image the horizontally left adjointable squares and horizontally left adjointable natural transformations.

(2) Dually for $(l \times \mathrm{id})^*$

(3) $(l \times r)^*$ similarly with image biadjointable squares and nat. transf.

Idea. (1) and (2) follow from the universal property of ADJ.

(3) in particular shows that biadjointability is symmetric.

$r^* : \mathrm{FUN}(\mathrm{ADJ}, \mathbb{D}) \to \mathrm{FUN}([1], \mathbb{D})$ $\quad$ $l^* : \mathrm{FUN}(\mathrm{ADJ}, \mathrm{FUN}^{\mathrm{ladj}}([1], \mathbb{D})) \to \mathrm{FUN}([1] \times [1], \mathbb{D})$

$\rightsquigarrow$ Image $\mathrm{FUN}^{\mathrm{ladj}}([1], \mathbb{D})$ $\quad$ $\rightsquigarrow$ Image biadjointable squares $\square$

IDEA OF THE PROOF OF THE LEMMA 1

$\forall (\mathbb{C}, Q_L, Q_R, Q_B) \in \mathrm{BADJDATA}_2$

$(-)_{\mathrm{badj}}$ admits a left adjoint $\iff$ $\exists\, \widehat{\mathbb{B}}(\mathbb{C})$ 2-cat and $h : \mathbb{C} \to \widehat{\mathbb{B}}(\mathbb{C})$ s.t.

$$\mathrm{FUN}(\widehat{\mathbb{B}}(\mathbb{C}), \mathbb{D}) \xrightarrow{\simeq} \mathrm{FUN}((\mathbb{C}, Q_L, Q_R, Q_B), \mathbb{D}_{\mathrm{badj}})$$

We construct $\widehat{\mathbb{B}}(\mathbb{C})$ and h in three steps:

(1)

$$\begin{tikzcd} \coprod_{\pi_0(Q_L)} ([1]\times[1]) \arrow[r, "\amalg(\mathrm{id}\times\eta)"] \arrow[d] & \coprod_{\pi_0(Q_L)} ([1]\times \mathrm{ADJ}) \arrow[d] \\ \mathbb{C} \arrow[r] & \mathbb{C}_L \arrow[ul, phantom, "\ulcorner", very near start] \end{tikzcd}$$

$\mathrm{Fun}(-, \mathbb{D}) \rightsquigarrow$

By previous lemma, $\mathrm{FUN}(\mathbb{C}_L, \mathbb{D}) \to \mathrm{FUN}(\mathbb{C}, \mathbb{D})$ is locally full inclusion with image the functors mapping Q to hor. left adj. sq.

(2) Same for $Q_R \rightsquigarrow \mathbb{C}_L \to \mathbb{C}_{L,R}$

(3) Same for Q_B, with

$$\coprod_{\pi_0(Q_B)} ([1]\times[1]) \xrightarrow{\amalg(\ell\times\eta)} \coprod_{\pi_0(Q_B)} (\mathrm{ADJ}\times\mathrm{ADJ})$$

$\rightsquigarrow \mathbb{C}_{L,R} \to \widehat{\mathbb{B}}(\mathbb{C})$

Then $h : \mathbb{C} \to \mathbb{C}_L \to \mathbb{C}_{L,R} \to \widehat{\mathbb{B}}(\mathbb{C})$

$\square$

PROOF OF THE RECOGNITION PRINCIPLE

Theorem (Recognition principle). An $(\mathcal{I},\mathcal{P})$-biadjointable functor $F : \mathcal{C}^{op} \longrightarrow \mathbb{B}$ is universal $\iff$ it is essentially surjective and for all $a \in \mathcal{C}$
$\mathrm{HOM}(F(a), F(-)) : \mathcal{C}^{op} \longrightarrow \mathcal{CAT}$ is a free $(\mathcal{I},\mathcal{P})$-biadjointable functor on $\mathrm{id}_{F(a)}$

Proof plan:

(1) $\mathrm{HOM}(F(a), F(-)) : \mathcal{C}^{op} \longrightarrow \mathcal{CAT}$ is a free on a $(\mathcal{I},\mathcal{P})$-biadjointable functor

(2) Any universal $(\mathcal{I},\mathcal{P})$-biadjointable functor $F : \mathcal{C}^{op} \longrightarrow \mathbb{B}$ is essentially surj.

(3) If $F : \mathcal{C}^{op} \longrightarrow \mathbb{B}$ essentially surjective and for all $a \in \mathcal{C}$

$$\mathrm{HOM}(F(a), F(-)) : \mathcal{C}^{op} \longrightarrow \mathcal{CAT}$$

is a free $(\mathcal{I},\mathcal{P})$-biadjointable functor on $\mathrm{id}_{F(a)}$, then F is universal.

Proof of the Recognition Principle

(1) $\mathrm{HOM}_{\mathbb{B}}(F(a), F(-)) : \mathcal{C}^{op} \to \mathrm{CAT}$ is a free on a (Σ, P)-biadjointable functor

It can be written as the composition $\mathcal{C}^{op} \xrightarrow{F} \mathbb{B} \xrightarrow{\mathrm{HOM}(F(a),-)} \mathrm{CAT}$

$\Rightarrow$ it is (Σ, P)-biadjointable

$G : \mathcal{C}^{op} \to \mathrm{CAT}$ (Σ, P)-biadjointable $\rightsquigarrow$ $\exists\, \overline{G} : \mathbb{B} \to \mathrm{CAT}$ and a diagram

$$\begin{tikzcd}
\mathrm{HOM}(\mathrm{HOM}(F(a),-), \overline{G}) \arrow[r, "ev_{id_{F(a)}}", "\sim"'] \arrow[d, "\sim"', "\widehat{F^*}"] & \overline{G}(F(a)) \arrow[d, equal] \\
\mathrm{HOM}(\mathrm{HOM}(F(a), F(-)), G) \arrow[r, "ev_{id_{F(a)}}"] & G(a)
\end{tikzcd}$$

$\Rightarrow ev_{id_{F(a)}}$ is an equivalence

- The top map is an equivalence by 2-cat Yoneda
- $\widehat{F^*}$ is an equivalence induced by $F^* : \mathrm{FUN}(\mathbb{B}, \mathrm{CAT}) \xrightarrow{\sim} \mathrm{FUN}_{(\Sigma,P)\text{-}badj}(\mathcal{C}^{op}, \mathrm{CAT})$

Proof of the Recognition Principle

(2) Any universal $(\mathcal{I},\mathcal{P})$-biadjointable functor $F: \mathcal{C}^{op} \longrightarrow \mathbb{B}$ is essentially surj.

$\mathbb{I} \subset \mathbb{B}$ full 2-subcat spanned by objects in the image of F

$\rightsquigarrow \hat{F}: \mathcal{C}^{op} \longrightarrow \mathbb{I}$ is $(\mathcal{I},\mathcal{P})$-biadjointable

$\rightsquigarrow \hat{F}$ admits an extension to $\mathbb{B}$ s.t.

$$\begin{tikzcd} & \mathcal{C}^{op} \arrow[dl, bend right, "F"'] \arrow[d, "\hat{F}"] \arrow[dr, bend left, "F"] & \\ \mathbb{B} \arrow[r, dashed] & \mathbb{I} \arrow[r, hook] & \mathbb{B} \end{tikzcd}$$

commutes

$\rightsquigarrow$ By the universal property of $\mathbb{B}$, $\mathbb{B} \dashrightarrow \mathbb{I} \hookrightarrow \mathbb{B}$ is the identity

$\rightsquigarrow \mathbb{I} \hookrightarrow \mathbb{B}$ is essentially surjective $\rightsquigarrow F$ is essentially surjective

PROOF OF THE RECOGNITION PRINCIPLE

(3) If $F : C^{op} \to \mathbb{B}$ essentially surjective and for all $a \in C$

$$HOM(F(a), F(-)) : C^{op} \to CAT$$

is a free (I,P)-biadjointable functor on $id_{F(a)}$, then F is universal.

By the universal property of $h : C^{op} \to \mathbb{B}_{I,P}(C)$, F extends to $\bar{F} : \mathbb{B}_{I,P}(C) \to \mathbb{B}$

It will suffice to show that $\bar{F}$ is an equivalence

Since $F = \bar{F} \circ h$ is essentially surj., $\bar{F}$ is essentially surj.

We want $\bar{F} : HOM(X, Y) \to HOM(\bar{F}(X), \bar{F}(Y))$ is an equivalence

(2) h essentially surj., it suffices $\bar{F} : HOM(h(a), h(-)) \to HOM(F(a), F(-))$ is an equivalence

$$\left.\begin{array}{l} (1)\ HOM(h(a), h(-)) \text{ free on } id_{h(a)} \\ HOM(F(a), F(-)) \text{ free on } id_{F(a)} \end{array}\right\} \rightsquigarrow \bar{F} \text{ is an equivalence}$$

$\bar{F}(id_{h(a)}) = id_{F(a)}$

$\square$